DATA CLEANING

1. Check for the number of missing values marked with '?' in each column of the “Medical Examinations” Table and "Hospitalization Details" Table.

   I have used formula =COUNTIF(A:I,"?") in Hospitalisation details.
   Ans = 15
   I have used the formula =COUNTIF(A:H,"?") in medical examination sheet.
   Ans = 2

2. Fill in the missing values of ‘month’ with Sep and ‘year’ with its average rounded to the nearest integer.

   Filling the missing values of ‘month’ with Sep :
   Applied a filter to the month column, select ?, type Sep in the first selected cell, select all the ? cells, and press Ctrl + Enter to fill them all with Sep.

   Filling the missing values of ‘year’ with its average rounded to the nearest integer:
   I have used the formula, =ROUND(AVERAGE(B:B),0) to find average year rounded to the nearest integer. Ans = 1983
   Then, Filter the year column for ?, enter the calculated average rounded to the nearest integer in the first selected cell, select all the ? cells, and press Ctrl + Enter to fill them all at once.

3. Determine the most frequently occurring values in the ‘smoker’, 'Hospital tier' and 'City tier' columns, and fill in the missing values accordingly.

   Most frequently occurring values in the ‘smoker’:
   Used Formula, =COUNTIF(H:H,"Yes") and =COUNTIF(H:H,"No").
   Ans: Yes = 488. No = 1845.
   Applied a filter to the month column, select ?, type "No" in the first selected cell, select all the ? cells, and press Ctrl + Enter to fill them all with No.

   Most frequently occurring values in the ‘Hospital tier’:
   Used Formula, =COUNTIF(G:G,"tier - 1") , =COUNTIF(G:G,"tier - 2") and =COUNTIF(G:G,"tier - 3")
   Ans: Tier – 1 = 309, tier – 2 = 1339, tier – 3 = 694.
   Applied a filter to the month column, select ?, type "tier - 2" in the first selected cell, select all the ? cells, and press Ctrl + Enter to fill them all with tier - 2.

Most frequently occurring values in the ‘City tier’:
Used Formula, =COUNTIF(H:H,"tier - 1") , =COUNTIF(H:H,"tier - 2") and =COUNTIF(H:H,"tier - 3")
Ans: Tier – 1 = 734, tier – 2 = 812, tier – 3 = 796.
Applied a filter to the month column, select ?, type "tier - 2" in the first selected cell, select all the ? cells, and press Ctrl + Enter to fill them all with tier - 2.

4. If any 'State ID' values are missing, consider filling them with 'Unknown' or using another appropriate strategy:

I filtered the State ID column for values marked with ? and replaced the missing values with Unknown. I then selected all the missing-value cells and used Ctrl + Enter to fill them at once.

Data Transformation:

1. Split the ‘names’ column in the “Customer Names” Table into 3 meaningful columns: ‘Title’, ‘First Name’, and ‘Last Name’.
   Title = Used Flash fills
   Last Name: Used formula =TRIM(LEFT(B2,FIND(",",B2)-1))
   First name: used formula =LET(x,SUBSTITUTE(B2,CHAR(160)," "),y,TRIM(MID(x,FIND(C2&".",x)+LEN(C2)+1,999)),TEXTBEFORE(y&" "," "))

2. Convert the "NumberOfMajorSurgeries" column in the “Medical Examinations” Table to numerical data by replacing non-numeric characters with meaningful numerical values.
   I used the Filter option on the `NumberOfMajorSurgeries` column to identify the non-numeric value “No major surgeries.” Since this indicates that the patient had zero major surgeries, I replaced “No major surgeries” with `0` and filled the corresponding cells. This converted the text entry into a meaningful numerical value while retaining the existing numeric values in the column.

3. Check for inconsistencies in the 'Heart Issues' and 'smoker' columns and propose corrective actions if necessary.
   I applied the Filter option to the `Heart Issues` and `smoker` columns and checked the available values. Both columns contained only `Yes` and `No`,

indicating that the data is consistent. Therefore, no corrective action was required.

4. Create a new column named “Weight Status” that categorizes BMI into different categories as below:
   I have used the formula =IF(A2<18.5,"Underweight",IF(A2<25,"Normal Weight",IF(A2<30,"Overweight","Obesity"))) to create column Weight status.

5. Create a new column named “Diabetes Status” and fill it as per the information given below:
   I have used the formula
   =IF(C2<5.7,"Normal",IF(C2<6.5,"Prediabetes","Diabetes")) to create column Diabetes.

6. Merge ‘year’, ‘month’ and ‘date’ columns in the “Hospitalization Details” Table into one column named ‘Date of Birth’ and format it in ‘DD-MMM-YYYY’ custom format.
   I have used the formula, =DATE(B2,MONTH(DATEVALUE(C2&" 1")),D2)
   Then, I selected the new Date of Birth column, pressed Ctrl + 1, selected Custom under the Number options, entered `dd-mmm-yyyy` in the Type box, and clicked OK to apply the required date format.

7. Calculate the ‘Age’ of each customer based on their ‘Date of Birth’ and the date of collection of the dataset, which is 8 th June 2023.
   I have used the formula =DATEDIF(J2,DATE(2023,6,8),"Y").

8. Format ‘charges’ column as currency ($)
   I selected the **Charges** column, pressed **Ctrl + 1** to open the **Format Cells** dialog box, selected **Currency**, chose **$ English (United States)** under Symbol, set the decimal places to **2**, and clicked **OK** to apply the currency format.

<u>Data Exploration, Analysis & Visualization:</u>

➢ Create a new sheet named “Healthcare", combine all three tables into one, using Customer ID as the common column, utilizing VLOOKUP.

➢ Retain the following necessary columns: Customer ID, First Name, BMI, HBA1C, Heart Issues, Any Transplants, Cancer history, NumberOfMajorSurgeries, smoker, Weight Status, Diabetes Status, Date of Birth, charges, Hospital tier, City tier, State ID, Age.

I created a new sheet named Healthcare and used Customer ID as the common column to combine the three tables. I entered the VLOOKUP formula using Customer ID as the lookup value and selected the appropriate source table and column number to retrieve the required information. I then filled the formula down for all customers to populate the large dataset. For the Date of Birth column, I formatted the retrieved values as `dd-mmm-yyyy` to display them correctly as dates.

VLOOKUP formula: =VLOOKUP($A2,'Table Name'!$A:$Z,column_number,FALSE)

Analysis using Pie/Donut Chart:

➢ What is the distribution of cancer history among smokers and non-smokers?

I created a PivotTable using Smoker as the filter, Cancer History as the category, and Customer ID as the count value. I then created separate Pie Charts for smokers and non-smokers to show the distribution of patients with and without a history of cancer.

➢ How does the total number of major surgeries and average HbA1C differ between patients with and without a history of transplants?

I used a Pivot Table to compare patients based on their transplant history. I placed Transplant History in the Rows section and Major Surgeries in the Values section, summarizing it using Sum to calculate the total number of major surgeries. I then added HbA1C to the Values section and changed its calculation to Average. This allowed me to compare the total major surgeries and average HbA1C between patients with and without a history of transplants.

➢ How do healthcare charges vary based on different weight statuses and diabetes statuses?

I used a Pivot Table to analyze how healthcare charges vary according to weight and diabetes status. I placed Weight Status in the Rows section and Diabetes Status in the Columns section. Then, I added Healthcare Charges to the Values section and changed the calculation from Sum to Average using Value Field Settings. This produced the average healthcare charges for each combination of weight status and diabetes status, making it easy to compare the differences between the groups.

➢ Can you compare the average charges for each hospital tier within different states?

I created a Pivot Table to compare the average healthcare charges for different hospital tiers across states. I placed State in the Rows section, Hospital Tier in the Columns section, and Healthcare Charges in the Values section. I then changed the calculation from Sum to Average using Value Field Settings. This allowed me to compare the average healthcare charges of each hospital tier across different states.

➢ Is there any correlation between age and both BMI and HbA1C in the dataset?

I used scatter plots to analyze the correlation between Age and BMI and between Age and HbA1C. I plotted Age on the X-axis and the respective variable, BMI or HbA1C, on the Y-axis. I then added a linear trendline to each scatter plot to observe the direction and strength of the relationship. This helped me determine whether BMI and HbA1C tend to increase or decrease as age changes.

➢ Explore the relationship between age and healthcare charges.

I used a line graph to explore the relationship between age and healthcare charges. I plotted Age on the X-axis and Healthcare Charges on the Y-axis. The line graph helped me observe how healthcare charges change across different age groups and identify any overall increasing or decreasing trend.

Dashboard Creation:

➢ Build an interactive dashboard that consolidates all key insights using the above visualizations. Ensure visual clarity and ease of interpretation for all chart types.

➢ Add slicers for the fields “Weight Status” and “Diabetes Status” to enable filtering across all visualizations, supporting comparison of health outcomes and charges based on body weight and diabetes condition.

I created an interactive dashboard in Excel by consolidating all the previously created visualizations into a single Dashboard sheet. I arranged the charts clearly, added suitable titles, and maintained consistent formatting to make the insights easy to understand. I then added slicers for **Weight Status** and **Diabetes Status** and

connected them to the relevant PivotTables and visualizations using **Report Connections**. This allowed me to filter the dashboard based on different weight and diabetes conditions and compare the resulting health outcomes and healthcare charges interactively.